



GE VERNOVA

ADMS

16.25.1

Glossary of Terms

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1. ADMS Terms

This chapter provides detailed definitions of the major terms used in relation to ADMS.

1.1. A

Active Display

The display currently visible in the active viewport. Also called the “current display”. When more than one display is visible in the current viewport, the display with input focus is the active display.

Active Viewport

The viewport currently on the screen that is affected by a given command. This is generally the viewport on the screen that has input focus. If no viewport has input focus, the active viewport is the viewport that last had input focus.

Active Window

The window on the screen that currently has input focus, as indicated by the window’s title bar being highlighted. Used in this sense, the term “window” can refer to a Browser viewport, an editing window in Display Builder, or any other Microsoft Windows application.

American National Standards Institute (ANSI)

The American association that publishes electrical standards.

American Standard Code for Information Interchange (ASCII)

A set of 7-bit binary numbers representing the United States (English) alphabet, punctuation, numbers, and other special symbols used in text representation and communications protocol.

See also [ISO-8859-1](#).

AMI

The Automated Metering Infrastructure for collecting outage, restoration, and meter information data for customer devices in the field.

AMI Report

An AMI report is treated similarly to a call, except that there is no possibility for priority information or comments. In addition, an AMI report can indicate that power has been restored, where there is no confirmation for restoration after a call (i.e., the customer does not call back to report that the power is back on again).

See also [Call](#).

AMR

Automatic Meter Reading.

Area Note

An area note represents a case note on a higher-level region, such as a city or a service center. These are used when it is known that there are widespread outages.

See also [Case Note](#).

Area of Responsibility

Areas of Responsibility (AORs) are used by the system to manage the portion of the network that each user is expected to control at any one time. An Area of Responsibility is associated with a specific device. An operator may be responsible for monitoring multiple Areas of Responsibility.

ARMS

The Asset and Resource Management System that provides the capability to manage service orders through the allocation and management of crews and resources.

Audible Alarm

A device that, upon receipt of a specified signal, produces a sound identifiable as notification of an abnormal event.

Automated Feeder Reconfiguration (AFR)

The AFR function provides plans to optimally restore service to customers, unload overloaded segments, and optimize the location of normally open points. The function can be used to reduce overloads on both substation transformers, or feeder segments. The function takes into account peak load conditions within a user-configured time window.

Similar to the other network optimization functions, AFR supports a number of problems formulations that define one or more operational objectives, including minimizing the number of switching operations, improving reliability, reducing overloads and voltage violations, and minimizing power losses. The AFR functions can be manually executed in real-time or study mode using the Network Solutions pop-up. The AFR output provides a set of recommended switching actions to the operator.

Automatic Execution of a Switching Order

Automatic Execution of a Switching Order means that the operator selects Execute and all the steps are executed without any further Operator intervention. In order for a Switching Order to be executed automatically, all the steps must be SCADA operations.

1.2. B

Boolean

A data type consisting of two constant values: TRUE and FALSE.
Boolean values are the result of testing relationships for truth and validity.

Branch

A line or transformer.

Browser

GE's standard user interface environment. Browser is a web-based browser that provides navigation to Distribution Management System (DMS) views, real-time Supervisory Control and Data Acquisition (Scada) displays, archive forms, and any Web-enabled applications.

Browser is made up of several components, including the Browser client, the Browser data server, Display Builder, the RUSER server, the replication data server, and the data entry server.

Browser Client

The graphical user interface component of Browser. The primary functions of the client are to retrieve and interpret display files, and to populate the resulting graphical representation of the display with live data from the Browser data server. The client also handles numerous related tasks, including command processing, managing the connection(s) to one or more data sources, message logging, and more.

Browser Viewer

Browser viewer is a form of the Browser client enhanced and optimized for control center operations. It has the same display call-up, command processing, and connection management capabilities provided by all versions of the client, while also offering many additional features that support the high availability, high demand, and high configurability needed in a production system.

Features provided by the viewer include:

- Multiple viewports and multiple views within a viewport
- User-definable toolbars
- Alarm viewer
- Charts and trending
- Display directory
- Rooms
- Support for multi-headed workstations (multiple monitors)

See also [Viewport](#), [Room](#).

Bus

An electrical bus is a junction, a point where several electrical circuits come together and are joined or connected. An electric substation is typically represented as one or more busses. Physically, a bus is usually a steel or aluminum tube or beam (there are actually three tubes or beams, one for each phase).

1. A section of a power system connected by circuit breakers, switches, or lines.
2. A conductor or group of conductors that serve as a common connection for two or more electric circuits within a station.

Bus Load Allocation

The Bus Load Allocation (BLA) function allocates loads (kW and kVAR) to distribution load nodes

that do not have telemetry, automatically accounting for other flow measurements to ensure consistency between allocated and metered loads.

The BLA procedure uses the real-time measurements from Scada and pre-defined customer type information to scale the distribution system loads to the real-time measurements. The expected value at a given date/time is obtained by multiplying the nominal load by a scaling factor that reflects the energy usage at a specific time. The nominal load values are determined from historical energy usage data. The allocation procedure requires minimum iteration with the power flow to maximize performance.

Button

1. An area or object in a display where, when the cursor is positioned there, specific keystrokes or mouse button actions execute specific commands.
2. A labeled, rectangular area of a dialog box that is used to control the information in that dialog box. Typical dialog box buttons are OK, Apply, and Cancel.

1.3. C

CAIDI

The Customer Average Interruption Duration Index (CAIDI) is calculated as the sum of all customer interruption durations (SAIDI) divided by the total number of customer interruptions (SAIFI).

Call

A call refers to one customer (a “normal” call) or zero customers (an “unassociated” call). A call generally indicates a power outage, but it can be for other reasons, such as a street light is out or the lights are blinking. A call can include potential priority information (for example, a wire is down) or comments from a customer.

A utility must specify mappings of trouble codes (what the problem is) and priority codes. Examples of trouble code mapping are 01 = Power Out, 02 = Street Light Out, 03 = Lights Blinked, etc.

See also [AMI Report](#).

Callbacks

There are two types of callbacks, where the system calls the customer as opposed to the customer calling the utility. The first type of callback is to verify that power has been restored; this callback can be generated by the operator when an incident is cleared. The second type of callback is to provide additional information to the customer while power is out (for example, the Estimated Time for Restoration (ETR) is later than first reported). These two types of callbacks are referred to as “restoration callbacks” and “case note callbacks”, respectively.

Callbacks are initiated by the ADMS, and sent to the IVR, to query as to whether the power is still out. If the power remains out, this results in additional outage calls being generated by the IVR. These call records can have all the original call message information.

Capability

1. (Black Start) The ability of a generating unit or station to go from a shutdown condition and start delivering power without assistance from the system.
2. (Operating) The maximum load-carrying ability of generating equipment or another electrical apparatus under specified conditions for a given interval.
3. (Transfer, Transfer Limit) An operating limit relating to the permissible power transfer between specified areas of the transmission system.

Capacity

The load-carrying ability of generating equipment.

Case Note

A case note refers to information that the system can provide to customers. It is made up of a coded reason (for example, code 200 represents “a crew has been dispatched to your outage”) and an ETR. Case notes are associated with outages (incidents); however, default case note information is also provided if the case note of an individual incident is not available.

Generally, the system maintains case notes on each service point, but it can be for an individual customer if an incident is smaller than a service point.

See also [Area Note](#).

Channel

A logical path connecting a user process to a physical device unit. A user process requests the operating system to assign a channel to a device so the process can transfer data to or from that device.

Channel Adapter

A channel adapter acts as a messaging client between the messaging system and an application that needs to communicate with the messaging system. A channel adapter can connect to the application at different layers of the application architecture, depending upon the need, and it can do so utilizing different communication mechanisms.

Character Field

A field that accepts any character string.

Check Box

A method of selecting or de-selecting an option used in dialog boxes and displays. A check box is a small, labeled square that is associated with a specific bit in a database. Clicking the left mouse button in the square toggles the option on or off, depending on its current state.

CIM

The Common Information Model (CIM) is an abstract model that represents all the major objects in an electric utility enterprise, typically involved in utility operations. The CIM provides a standard way of representing power system resources as object classes and attributes along with their relationships. This is accomplished by defining a common language (i.e., semantics and syntax). It is a schema, not a database or a specific data file format.

Circuit

A system of conductors and its component parts through which an electric current flows or is intended to flow.

Circuit Breaker (CB)

A device designed to trip or open a transmission circuit while under load (i.e., while carrying current). Comparable to a fuse or circuit breaker in the home. Because of the high voltage and potentially high current, it must be specially insulated. Usually, this is accomplished by immersing the contacts in a sealed tank of insulating oil or inert gas (SF₆) or by using a blast of high-pressure air to extinguish the electrical arc created when the two contacts are separated.

1. A switching device that automatically opens an electrical circuit when a predetermined current overload level is reached.

Circuit breakers, which can also be used to open and close circuits manually, are not normally damaged (as is a fuse, for example) when they activate.

2. A switching device capable of making, carrying, and breaking currents under normal circuit conditions, and also making and carrying for a specified time and breaking currents under specified abnormal conditions such as those of a short circuit.

Clause

See [Test Clause](#).

Clearance

Formal, scheduled, and recorded permission issued by the dispatcher for crews to work on specific equipment. Coordinated with switching and tagging operations to ensure that equipment is de-energized and safe for work.

See also [Switching Order](#).

Client Workstation

A workstation networked to a host CPU from which displays are accessed and viewed.

Closed Loop Operation Mode

In the Closed Loop operation mode, Fault Isolation and Service Restoration (FISR), Automatic Feeder Reconfiguration Function (AFR) and Load and Voltage/VAR Management (LVM) applications have the ability to automatically execute switching steps of remotely controllable devices such as breakers and capacitors.

In particular, FISR (used to isolate faults and provide optimal service restoration) and AFR (used to find a configuration which eliminates the overloads) create a switching plan that provides service restoration and, in the closed loop mode, the best switching plan is automatically selected and passed to Switching Operations.

Similarly, in the closed loop mode, LVM directly issues commands for the control of substation capacitors, feeder capacitors, Load Tap Changer (LTC) controllers, and voltage regulators for a pre-defined problem formulation.

Command Line

An enterable text box in which commands can be typed and issued.

Each viewport contains its own command line by default. The main viewer window also includes command lines on some of its tabs. Commands issued through a Browser command line can be recalled simply by using the combo box that drops down from a control on the right edge of the command line itself.

Conditional Test

An instruction to examine (test) database values to determine if they meet specific conditions. Conditional tests include at least one test statement, which is hierarchically composed of test clauses and test phrases.

Conditional Text

A placeable element that provides the capability to display different text strings, based on the outcome of a conditional test of database values.

Confirmed Outage

The dispatcher has marked the device as open in the Outage Management System (OMS).

Context-Sensitive Help

Context-sensitive help is the ability to have help information appear on the screen that is specific to the current information in an application window.

Contingency

1. An outage or change that occurs in the power system, such as the loss of a transmission line, a generator, a transformer, or any other equipment that goes out of service. Although generally predictable in their effects, contingencies nevertheless are not normal occurrences.
2. (Single Element) A contingency involving the loss of one element.
3. (Double Element) A contingency involving the loss of two elements.
4. (Limiting) A contingency that establishes the transfer capability.

Contingency Analysis (CA)

An application that studies the effects of contingencies on the power system and reports to the dispatcher or engineer the equipment limits that would be exceeded if the contingency occurred.

Control

SCADA for multiple purposes: master SCADA, communication front end plant gateway, substation server.

Crew

A crew represents a set of employees and vehicles that can be assigned to incidents. Crews can be built "on the fly" or initialized from a list imported into the system. A crew must have at least one employee (the contact employee for the crew) and it can contain from one to many vehicles.

There are potentially many identifiers for a crew. A Workforce Management (WFM) ID specifies a common ID for use in communicating to a third-party crew management system. A radio ID can be used for direct radio communication between the operator and the crew, and one or more telephone numbers can be specified. The IDs are defaulted from the employees and vehicles that are used to build the crew.

A crew:

- Has capabilities as supported by the employees and vehicles that make up the crew
- Can be marked as unavailable to the system, so that they remain in the system but are not shown when presenting the operators with a list of crews to assign
- Can be assigned to zero or more incidents
- Can only be dispatched to one incident at a time

Crew Assignment

Crew assignments are created by the system when an operator requests the assignment. A crew assignment has one pointer to an incident and one pointer to a crew. When an incident is completed and cleared, the assignment is automatically deleted by the system.

An interface to a WFM system deals with incidents and crew assignments. For example, a WFM can specify the status of a crew assignment, which is then applied to the crew and the incident.

It is possible for the crew assignment to be created and maintained by the work management system. That is, the ADMS can tell the WFM system about an incident, and the WFM dispatcher can assign a crew to it and inform the ADMS of which crew was assigned.

Current Display

The display currently visible in a viewport. Also called the “active display”.

Current State

The Static Model coupled with Dynamic Events equals the current state of the power network.

Cursor

An onscreen symbol that reflects the position of the mouse (or trackball). The cursor's shape often indicates the type of operation being performed.

See also [Text Entry Cursor](#).

Customer

Customers are created by the Customer Information System (CIS) and passed (in an offline interface) from the CIS. These are compiled into a customer file. The fields associated with a customer are all customer-defined, except that some fields are required:

- Service Point ID: This refers to a service point in DNOM, or the point in the network where the customer is connected.
- Account ID: This is a unique identifier for this customer.
- Premise ID: The AMI interface requires a premise ID, which can be the same as the account ID.

- **Meter ID:** For AMI communications, the meter ID is required.
- **Priority Indicator:** This indicates that the customer is critical for some reason — for example, the customer is a key customer.
- **Region 1 and Region 2:** This indicates the geographical location of the customer for reporting purposes. Examples of these regions are City and Service Center.
- **X,Y Location:** The geographical location is used to display the customer.

In addition to the static fields supplied by the CIS, the system maintains a number of real-time fields, such as the call state, the current AMI state, and a pointer to its active incident.

Customer Support System

The Customer Support System is used to collect trouble calls from customers.

1.4. D

Data Compression

The ability to compress data sent between the data server and client processes.

DDL File

Any ASCII file that contains a valid collection of Display Definition Language declarations. DDL files use the .ddl extension by default.

See also [Display Definition Language](#).

Declutter

The feature of simple displays that allows objects in a display to be removed from or added to the display at discrete zoom factors.

Declutter Zoom Factor

The zoom factor at which an object has been defined to be removed or added to a display. Each item defined to declutter is defined with a high and a low declutter zoom factor.

Default Zoom Level

The zoom factor at which a display has been defined to initially appear at call-up.

Dialog Box

A window that is used to solicit additional information from users through the use of list boxes, text boxes, toggles, check boxes, option buttons, sliders, or other buttons.

Display

See [Tabular Display](#) and [One-Line](#).

Display Definition Language (DDL)

A declarative, ASCII-based programming language for defining elements and displays. Element and display definitions are saved in files called “element sets” and “display sets”, respectively. Element set and display set files are commonly referred to as “DDL files”.

DDL files are created using the Display Builder, which is a WYSIWYG editor for the Display Definition Language.

Display Name

The name given to a display in its definition. This name is used to call the display to the workstation, both in the online environment and in the Display Builder editing environment.

Display Set

A named group of related displays and picgroups.

Usually, this is the station internals one-line display.

See also [Display Definition Language](#).

Display Title

A text string, up to 80 characters in length, used to describe a display to the user. It is defined in the Display Properties dialog box in the Display Builder. The display title appears in the viewport title bar when the display is called into a viewport.

DNAF

The Distribution Network Analysis Functions (DNAF) form an integrated package of applications designed to analyze and optimize the network. The memory resident Distribution Network Object Model (DNOM) provides the connectivity and attributes data required by the DNAF.

The DNAF can be executed in both real-time and study modes. The real-time mode uses the current power system conditions (SCADA measurements, capacitor time of operation schedules, load schedules) in evaluating the system. The study mode solves operational studies using data retrieved from a saved case or a real-time snapshot of the system. The future operating point for studies is estimated using typical load shapes. The load shape information is based on the study time and day type.

DNOM

The underlying infrastructure of ADMS is the dynamic Distribution Network Object Model (DNOM).

DPF & DPFIO

Distribution Power Flow. DPFIO is the interface object between DNAF functions and UPFIO objects.

The core network analysis engine is the Distribution Power Flow (DPF). The DPF finds a phase domain solution to unbalanced power system network models, including the individual bus phase voltages, phase angles, and branch device flows. The DPF is capable of modeling and solving any unbalanced distribution system configurations, including networked (meshed) systems and radial systems.

Drag and Drop

The procedure of initiating a display call-up in one window or viewport, and transferring completion of the command to another viewport, by means of a click-and-move action of the mouse. The objects from which displays can be dragged include display buttons, drop-down and

pop-up menus, and the display directory window.

Drop-Down Menu

A menu that appears in the menu bar of a viewport, and is activated (dropped down) by pressing the left mouse button on the menu's title in the menu bar.

See also [Pop-Up Menu](#).

Dynamic Data

Dynamic data refers to all actions and events that are applied on top of the static description of the network to reflect the current state of the distribution grid. This dynamic data includes:

- SCADA measurements
- Operator actions on non-telemetered devices
- Operator actions related to tags and annotations
- Operator actions related to cuts, jumpers, and grounds

1.5. E

Electrical Address

Location on the network where a customer is connected, typically a distribution transformer.

Energize

To provide energy supply to a piece of equipment.

Energy Management System (EMS)

A collection of applications, computers, and Scada equipment used to monitor and control a power generation and transmission system. Used interchangeably with the term "power management system".

Entry Substation

The substation from which the DNAF functions are triggered. For example, when you try to solve the DPF for the RAVEN substation, RAVEN is the entry substation. For applications that include adjacent substations in the calculation, such as FISR, the entry substation (for example, RAVEN) and its adjacent substations are considered together.

ETR

While the incident is active, it will have an Estimated Time of Restoration (ETR). The system automatically calculates this ETR based on the reason for the outage, the area, whether the equipment to be repaired is accessible, and activity on the system (for example, if there is a major storm in progress). The ETR may also be set by the operator.

External

In the context of the ADMS Substation Editor, this term refers to the distribution network equipment that is normally fed by the substation but is located outside the fence.

See also [Internal](#).

1.6. F

Failover

The term is used to describe the event of failing over — that is, when system operations shift from one set of computing equipment to another in a system specifically designed to make this possible. It is initiated automatically by the Configuration Manager when equipment or software fails.

Fault

Reaction to a disturbance which can be sporadic, permanent, semi-permanent, self-extinguishing, or random. Examples:

1. A physical condition that results in the failure of a component to transmit electrical power in the manner for which it was designed.
2. (Permanent) A fault that prevents the affected device from being returned to service until physical actions are taken to make repairs or to remove the cause of the fault.
3. (Transient) A fault that occurs for a short or limited time, or that disappears when the faulted device is separated from all electrical sources, and that does not require repairs to be made before the device can be returned to service either manually or automatically.

Fault Isolation and Service Restoration (FISR)

The Fault Isolation and Service Restoration function (FISR) is designed to optimally restore service to customers. In the case of a sustained outage, FISR will create a Switching Plan that isolates the faulted area and provides service restoration by using optimal feeder reconfigurations, including cascading reconfiguration to off-load one feeder in order to pick up part of the load of an adjacent feeder.

FISR utilizes a problem formulation that defines one or more operational objectives, including minimizing un-served kW, minimizing customer minutes interrupted, minimizing the number of switching operations, and minimizing voltage drop along the reconfigured feeders. The problem formulation also specifies the network and operational constraints, including segment overloads, voltage violations, and relay settings.

The FISR function can run in two modes:

- Closed Loop Mode: The best plan is automatically selected and executed. Obviously this means that only SCADA controlled switches can be included in Closed Loop plans.
- Advisory Mode: The plans are presented to the operator in ranked order. The selected plan can then be sent to Switching Operations to automatically create a formal Switching Order.

Feeder

1. All circuit conductors between the service entrance and the final branch-circuit overcurrent device. For example, circuit breakers.
2. A distribution circuit, usually 50 kV or less, that supplies customer load.

Future Equipment

Electrical equipment that is not yet part of the network. Once installed and commissioned, the equipment will transition from the Future state to the Active state and be usable by the system.

Future Remove Equipment

Electrical equipment that is planned for removal from the network. A device in this state is still in use on the network, but once decommissioned, will no longer be used by the system. After decommissioning, the equipment will transition from the Future Remove state to the Removed state.

1.7. G

GAB

See [Graphic Attribute Block/Bundle](#).

GIS

The generic term used for "Geographic Information System", the GIS provides the map data for the ADMS.

"Go Back Switch Order"

The system has the ability to generate a "Go Back" switch order to go back to the network configuration that was in place at the start of the Switch Order. This may not be a return to the normal configuration.

See also ["Return to Normal" Switch Order](#).

Graphic Attribute Block/Bundle (GAB)

A collection of visual attributes applied to a display primitive or element. A GAB can be given a name and saved in an element set; this is called a "named GAB". Alternately, individual primitives can have their graphic attributes set independently of all other primitives and elements; this is called a "private GAB".

Ground

A conducting connection (whether intentional or accidental) by which an electric circuit or equipment is connected to the Earth or to some other conductive body large enough to serve in place of the Earth.

Ground Loop

A condition (normally undesirable) where the grounds of energy-generating or energy-converting devices are connected and varying voltage potentials cause a current path within the ground itself.

This can occur, for instance, when different voltage potentials cause a current path through a building's structural steel, which is serving as a ground.

Ground Return

The portion of a power transmission system in which one of the conductors is replaced by the ground.

Grounded

Connected to the Earth or to some extended conducting body that serves instead of the Earth, whether the connection is intentional or accidental.

Grounding Conductor

Any path of copper or other suitable metal arranged in such a way as to connect electrical equipment to the ground electrode.

1.8. H

Help Menu

The standard viewport menu item used to access Help files by application, task, or display.

1.9. I

Icon

A generally compact, graphic representation of an application, object, or function.

In Service

A term used to indicate that a component or element is connected to the system and fulfilling its function.

See also [Outage](#).

Incident

An incident represents an occurrence on the power system that needs to be repaired, and generally indicates that customers have an outage.

There are several types of incidents: single customer incident, predicted incident, momentary incident, confirmed incident, manually confirmed incident, isolated incident, and planned incident. For more information, refer to the *Outage Management System User Guide*.

Input Focus

The ability of a window (application window or text window) to echo keystrokes and accept commands. Only one window per screen, and only one text entry cursor per window, can have input focus at any given time.

Internal

In the context of the ADMS Substation Editor, this term refers to the equipment that is part of the substation (i.e., located inside the fence).

See also [External](#).

IO Object

IO objects are temporarily created input/output objects to access the DNAF functions. Before invoking the DNAF functions, some IO objects must be created. They provide input for analysis, and DNAF stores the results in the IO objects. Finally, the results are transferred from IO objects to DNOM objects. After that, IO objects are deleted.

ISO-8859-1 (also known as Latin-1)

ISO-8859-1 is an 8-bit character set that is used to represent characters.

ISO-8859-1 is one of a collection of 8-bit character sets that are used to represent most of the world's non-ideographic character sets. ISO-8859-1 is commonly referred to as "Latin-1", and it is used to represent Western European languages as well as English. The first 128 values of Latin-1 are identical to ASCII. Most programs and operating systems today use Latin-1 as the default character set rather than ASCII.

IVR

The external Interactive Voice Response system utilized for automatically collecting trouble call information from customers. Example: "Press 1 to report a power outage, press 2 to report an emergency".

1.10. L

Label

A text string that describes the use, function, and contents of dialog box mechanisms such as text boxes, toggles, menus, etc.

Layer

Discrete units of a display that act like individual overhead transparencies. They can be viewed independently or as a group, with objects on the upper layers occluding objects on the lower layers when they appear at the same coordinate location, while allowing non-occluded objects on the lower layers to be visible.

Limit Monitor (LIM)

The limit monitor (LIM) function checks for branch flows and voltages that are outside of acceptable operating limits. The LIM function identifies all limit violations based upon the most recent DPF solution. All load (and station bus) voltages and line flows calculated by the DPF are checked against their limits.

Load and Volt/Var Management

Load and Volt/VAR Management automatically controls capacitors and voltage regulation devices on the distribution feeders and in the substations to achieve multiple objectives, namely Demand Reduction (1-5%) in Energy Conservation mode or Peak Shaving mode as well as Loss Minimization/Power Factor management. By using analysis of a significant portion of the distribution network and coordinating the operating commands for all active regulating devices,

LVM achieves a considerable improvement over conventional load and volt/var control techniques.

1.11. M

Manual Execution of Switching Steps

Manual execution of switch steps enables the Operator to manually initiate the execution of each step of the Switching Order, acknowledging its successful completion, and then proceeding to the next step. The individual steps are considered to be manual because the field staff are performing the switching.

Mapboard

A schematic of the power system used by power management systems to provide operators with a quickly understandable view of the system and its status. Some mapboards are static: a simple schematic of the power system, with tags and pins (placed on the mapboard by operators) providing pertinent information and updates. Others are dynamic: a schematic of the power system with display elements such as lights and light strips that are actuated by a mapboard interface to show status changes as they are detected by the Scada system.

Megavolt-Ampere (MVA)

A unit for measuring the complex or apparent power, expressed as the product of the voltage (in megavolts) and the current (in amperes).

Megavolt-Ampere Reactive (MVAR)

A unit for measuring reactive power.

Menu

A list of options (called “menu labels”) that a user can choose, each option causing some distinct action (command) to occur.

See also [Drop-Down Menu](#), [Pop-Up Menu](#).

Menu Bar

The area on a window that contains the names of the drop-down menus available for that window. Not all windows have a menu bar.

See also [Drop-Down Menu](#).

Menu Label

A text string that describes the available choices in a menu. Each label is associated with a command (including the command to bring up a submenu) that is executed when the label is selected.

Message Log

Messages logged by application programs. Usually called the “Log display”, although displays are not the only mechanism used to view the Message Log.

Momentary Incident or Outage

An outage that ends before the system-wide momentary timeout is reached (this timeout is utility-specific, typically between 30 and 300 seconds). Momentary incidents are not presented to the operator and are not included in the performance indices.

1.12. N

NETIO

Habitat's subsystem that supports interprocess communications.

NETIO includes several APIs that encapsulate all details associated with interprocess communications, including transport and operating system issues. In addition, NETIO makes use of a real-time name service so that tasks do not have to be aware of where remote parties are located.

Network Selection

The network selection is one process through which the analysis scope is selected for some DNAF functions. For different functions, there may be different analysis, meaning that some DNAF functions may have a different network selection. Network selection defines which network (or part of the network) is to be selected for analysis.

Network Topology

Network topology is the sum of the manual and SCADA changes to the switching devices on the network.

Example: If a feeder trips, SCADA tells the network server, which traces the event downstream to determines which customers are de-energized and creates an incident.

Nominal

The value you find in a specification.

1.13. O

Ohm

The unit of measurement for resistance in an electrical circuit.

On Application Help

An option on the standard Help menu that accesses the table of contents or index for the user guide, in which the basic functionality of the application of the current display is described.

On Display Help

An option on the standard Help menu that accesses a detailed description of the current display in the viewport.

On Task Help

An option on the standard Help menu that accesses a list of tasks that can be performed using this display. Users can then select the desired task from the list, to view detailed procedures for performing the task.

One-Line

A diagram or display of an area in the power system where one line on the diagram represents the three phases of one transmission line.

Online Documents

Product documentation viewed using a computer and monitor, as opposed to printed manuals.

Operator

The control center operator is the person operating a console in order to monitor and control the power network.

Out of Service

A term used to indicate that a component or element is disconnected from the system.

Outage

A set of customers whose power is out of service for the same reason.

Outage Calls (or Calls)

These are initiated by the power consumer to indicate that the power is out (or blinking, or dim lights, or there is an emergency). An outage call is normally associated with a single customer. The message flow is from the CSS, or from the IVR to the ADMS. Priority, unassociated calls are a special case of outage calls.

1.14. P

Panning

The ability of a user to reposition the viewport over a display by moving the cursor in the viewport.

Per Unit (PU)

A normalized set of units. For example, one “per-unit” volt is equivalent to the nominal voltage for a branch. Thus, on a 112-KV line, 1 PU volt is 112 KV. Similarly, frequency is normalized to 60 Hz (50 Hz in some countries) so that 1.0 PU Hz is 60 Hz, and power is normalized to 1 MW so that 50 PU MW is 50 MW.

Permission

The level of access granted to a given client session for displays and components of displays.

Permission Area

A category that has been created by a system analyst using an arbitrary name, under which permissions are granted. Using permission area tests, Display Builder can configure displays

and components of displays to be visible or not visible, depending on the console name used to call up the display.

See also [Permission Area Test](#).

Permission Area Test

Permission area tests are used to control operator interactions with applications and their databases, by limiting whether a display can be called up at a workstation. In a permission area test, the user defines the access permission area ID(s) used in the display-based comparison check with the PERMIT database, or specifies where the area ID is found.

Phase Mismatch

One phase is de-energized or is being supplied from the opposite end of the line.

Planned Outage Study (PLOS)

This function provides the operator with an automated way to prepare, evaluate, and manage planned outages. The aim of Planned Outage Study (PLOS) is to recommend switching plans to support planned outages.

POS is only available in Study Mode. The user specifies the network device that is to be taken out of service and POS will generate one or more possible switching plans.

Planned Switching

Planned switching is generally scheduled ahead of time and performed in context of maintenance or construction that must be performed by field employees in substations or along distribution lines. The field employees assigned to perform the switching operation generally receives printed copies of the switching orders in advance of the operation from a Control Center Operator who has permission to generate switching orders.

Platform

An energy management system solution, including SCADA, transmission operation, generation management.

Pop-Up Menu

A menu that is activated at the cursor's location as a result of a user action. Menus can be attached locally or as overall attachments. They appear when the cursor is placed on the item or display to which the menu is attached and the Menu command is issued.

A pop-up menu always appears at the cursor's current location in the display.

See also [Drop-Down Menu](#).

Power Factor

The ratio of active (real) power to apparent power, expressed in percent or a decimal number.

Power System

A group of one or more generating sources and/or connecting transmission lines operated under common management or supervision to supply load.

Power System Analyst

This person observes the state of the power network and performs “what if” contingency analysis. This person understands how the system is modeled, and the relationship of all the equipment in the model.

Predictive Outage

Based on customer calls, the system predicts the protective device that will open.

Primitives

Text and basic geometric shapes (rectangles, circles, arcs, ellipses, polygons, and polylines) that can be created in element or display scratch pad editors.

PROCMAN (Process Manager)

An application within Habitat that is responsible for the management of programs running in an Habitat-based system. PROCMAN deals with task startup, sequencing, scheduling, and failure detection.

Protection Validation

The Protection Validation (PRV) function identifies unprotected or improperly protected application protection areas and compares the fault currents with relay settings and the interruption capacity of breakers and reclosers. The Protection Validation (PRV) application performs three major functions:

- Breaker and recloser protection setting check.
- Protection presence.
- Protection coordination.

This function can calculate phase-to-ground, double-phase-to-ground, and three-phase fault currents for each application protection area, where an area is defined by feeder circuit breakers and field reclosers.

1.15. Q

Query

A structured request for the specific database values needed to populate a given layer of a display or display page.

The Browser client sends one or more queries to the Browser data server each time a display is called up. The data server resolves the query and sends the results — made up of database values and other real-time data — back to the client, where they are used to populate a given layer of the display. One query is processed for each layer of the display that contains refreshable data.

1.16. R

Reactance

The physical property of a circuit component to hinder the flow of alternating current.

Real Time

1. Literally, in the present.
2. A method of data processing in which data is acted upon immediately instead of being accumulated and processed in batches. A real-time system controls ongoing processes and delivers output (results, controls, etc.) within the time needed for effective control.

Record

1. A group of related fields that make up a database component.
2. A collection of data that describes a database component. Each individual piece of data is a field record address. The starting address of a record's field data in a record-oriented partition.

Relay

A device designated to detect an electrical fault (short circuit) or other abnormality on a transmission line or other power system element, and to initiate corrective action to protect the system, usually by removing (tripping) the faulted element.

Note

Some relays are designed to do significantly more.

Remote Terminal Unit (RTU)

The interface between power system field devices such as voltage meters and the Energy Management System (EMS). In a sense, a remote terminal unit is the first level of the EMS, inasmuch as it gathers information about actual power flow and use. This data is the raw material on which the EMS operates.

There are two types of RTU:

- **Field RTU:** Gathers data in the field (that is, at substations) for transmission to the energy management system's computers. Field RTUs normally use voice grade communication channels to transmit data upline.
- **Local RTU:** Installed at the same site as the power management system computers. Local RTUs can be identical to field RTUs in component equipment and type of data gathered, or they can be used to gather data about local frequency, temperature, humidity, and fire alarm status. Local RTUs generally need no telephone line-modem links because they communicate directly with the energy management system computers.

"Return to Normal" Switch Order

In many cases, after an unplanned outage, customers may be restored but there can still be a number of switches across multiple feeders that are left in abnormal states. Therefore, a

procedure is required to safely return the switching to its normal state while avoiding any further de-energization of customers.

The Feeder Reconfiguration (Return to Normal) function is used to develop this switching procedure and return power system circuits to their normal configuration. The output of the Return to Normal function is one or more switching plans. If the Return to Normal function has been run in study mode, the switching order can be fully evaluated prior to being transferred to real time.

See also ["Go Back Switch Order"](#) and [Automated Feeder Reconfiguration \(AFR\)](#) for more information about this function.

Room

A savable, named arrangement of viewports and views, and the displays associated with them that can be called to a workstation.

Rooms are supported in the Browser viewer only. They are not available when running the Browser client inside Microsoft Internet Explorer or as an embedded control.

See also [Browser Viewer](#).

Rubber-Band Selection

The method of selecting items for some operation (copy, move, delete, etc.) by enclosing the items in a user-sizable rectangle.

Rubber-Band Zoom

The method of zooming an area of a simple display to fit the current viewport size by enclosing the area in a user-sizable rectangle.

1.17. S

Savecase

An off-line copy of a dataset. A study environment, once set up, may be saved as a "Save Case".

SCADA

Supervisory Control and Data Acquisition. An application used by dispatchers to monitor and control the power system. The GE product name is "Scada".

For ADMS substations, Scada one-line displays for the management of the substations are shown in the same user environment removing any complications of navigating back and forth between dissimilar systems. Real-time SCADA data is passed to the Network Operations Model and is immediately displayed to all users via the network views. The toolbar button shows that the switch is controlled by SCADA and when clicked the SCADA control popup is displayed.

In the case of automated switches on the feeder, the SCADA controls are provided directly on the geographic network view. The toolbar button shows that the switch is controlled by SCADA and when clicked the SCADA control popup is displayed.

SCADAMOM

The database used by the Scada application.

Secondary Loads

System loads that are not classified as primary loads.

Service Group

A service group specifies a group of service points, for reporting and querying purposes. There are two automatically created types of service groups: feeder and substation. Additional service groups can be configured by supplying a set of polygons to the system, such that service points that fall geographically in a polygon are part of the service group. For example, operating centers and counties can be specified as service groups, so that it is possible to determine the number of customers out in each operating center and county.

Service Point

A service point is a DNOM object that specifies a node in the network where one or more customers are connected. This is generally the low side of a distribution transformer, but it can be any node on the network. In some cases, this may be a point on the primary (for a primary metered customer) or a point on the low voltage network if this is modeled.

The system maps customers to the network by service point. The service points are specified in DNOM, and each customer in the customer list supplied by the CIS supports the name of a service point. The customers must be remapped when the DNOM model changes or when the customer model changes.

Since a service point is a DNOM object, it has an X,Y geographical location as an attribute.

Short Circuit Analysis (SCA)

The Short Circuit Analysis (SCA) function calculates the maximum short circuit current available for various fault types. It automatically finds the highest impedance fault path in the dynamic network. The SCA function provides a level of Protection Validation that ensures all components are properly protected.

Smart Grid

The term “Smart Grid” refers to a modernization of the electricity delivery system so it monitors, protects, and automatically optimizes the operation of its interconnected elements – from the central and distributed generator through the high-voltage network and distribution system, to industrial users and building automation systems, to energy storage installations, and to end-user consumers and their thermostats, electric vehicles, appliances, and other household devices.

Static Data

Static data essentially refers to the description of the distribution network in its normal configuration, including equipment characteristics and connectivity. This data is stored in binary files, also called “station model files” because there is one file per station.

Status Bar

A narrow band at the bottom of the window that provides additional information.

Study Mode

Offline load flow, voltage estimation and fault level calculation based on network snapshots (what if analysis); checking of switching sheets.

Study Mode allows the Network View and Network Analysis and Optimization functions to be run on a client workstation. Study Mode uses the same static models as real-time, but with a separate instance of the dynamics. This removes the requirement to separately set-up and maintain a study environment. Changes made to the study mode instance of the network area are applied only to the operator's workstation and are not visible to others. Both real-time and study modes can be run concurrently on the same workstation.

Subsystem

1. In the context of the Energy Management System (EMS), the term "subsystem" refers to the Scada, Network Security and Optimization, Generation Control and Optimization, and Grid Simulation components that together make up an EMS. Each type of subsystem is in turn made up of numerous applications; for example, the Scada subsystem is made up of the SCADAMD, Scada, Load Shed, and Tagging.
2. The part of the network that is sensitive to a particular contingency. When Contingency Analysis (CA) initializes, a subsystem is built for each contingency. The subsystem includes all nodes that are sensitive to the components that the contingency outages. During local screening, the contingency's subsystem is the only part of the network that is searched for potentially harmful conditions.

Subtransmission System

Circuits that emanate from the bulk power source and supply distribution substations.

Supervisory Control

A form of remote control that includes an arrangement for the selective control of remotely located facilities by an electrical means over one or more common interconnecting channels.

Supervisory Control and Data Acquisition (SCADA)

See [SCADA](#).

Switchable Section

The portion of the electric power distribution network bounded by two or more feeder main switching devices (breaker, recloser, sectionalizer, disconnect, and other).

Switching Operations

Switching Operations is the formal process of performing switching changes on the network for the purposes of maintenance, network reconfiguration, and restoration of unplanned outages and augmentation of the distribution network. Switching Operations support the creation, verification/simulation, and execution of switching orders associated with planned and unplanned switching activities.

Switching Order

A Switching Order is a formal document that records all the actions necessary to achieve a required operation such as isolating a section of network and making it safe for maintenance

work. It documents the reasons for the work, instructions to be executed, and the detailed record of the execution process. A Switching Order may contain switching actions, tagging actions and other non-switching instructions necessary for the operation (for example placement of safety barriers). In most utilities the Switching Order is the formal record that all the required procedures have been carried out and therefore considered to be a legal document.

Switching Plan

A list of Switching Actions required to perform an operation such as isolating a section of the network. These plans are normally generated by an automated process such as FISIR or AFR.

Symbol

A placeable element that consists of one or more primitives (excluding text), used to represent common objects in a display.

1.18. T

Tabular Display

ADMS tabular displays are accessed from ADMS menus. They show information about the distribution network. ADMS tabulars can display both real-time and study mode results.

Tag

A tag is a physical (for example, field tag) or electronic marker on a switch device to prevent operation. A tag gives details of the clearance that is currently issued for the device.

Telemetry

1. Remote measuring device: a device used to record information about a remote object or event.
2. A device for measuring distances directly.
3. Transmit data: to collect and transmit data about a remote object, especially using a satellite.

Test Clause

The part of a conditional test that determines which database values are to be tested and what result constitutes a successful test. Test clauses are of the IF, ELSEIF, or ELSE form.

See also [Test Phrase](#), [Conditional Test](#).

Test Phrase

A secondary part of a conditional test that provides additional conditions for a test clause (i.e., additional database values to be tested and what result constitutes a successful test). Test phrases are of the AND (conjunction) or (disjunction) form.

See also [Test Phrase](#), [Conditional Test](#).

Text Box

A dialog box method of entering text. An area of the screen with a text entry cursor that accepts

the entry and manipulation of text strings when it has input focus.

See also [Text Entry Cursor](#).

Text Entry Cursor

A cursor in a text box (or window) that indicates where text is entered.

Title Bar

A narrow band at the top of a window that provides information about the document currently active in the window or viewport and about the commands for controlling (minimizing, maximizing, closing, etc.) the application.

Tooltip

A descriptive text box that is displayed when the mouse pointer is held over a button or display object.

Toolbar

A collection of buttons with associated commands that provides a convenient way to access the features of an application. A toolbar can also contain drop-down list boxes.

Topology Processor

A function of the RTNET application that studies the status of circuit breakers and determines how the power system is connected.

Transformer

1. An electromagnetic device for transferring energy from one circuit to another of different voltage levels in an alternating current system.
2. A device that advances or retards the phase angle relationship of one circuit with respect to another to control power flow.

Syn Phase Angle Regulator

3. (Voltage Regulating) A device that increases or decreases the voltage magnitude relationship of one circuit with respect to another. It is often used to control reactive power flow.

Transmission

In power system usage, the bulk transport of electricity from large generation centers over significant distances to interchanges with large industries and distribution networks of utilities.

1.19. U

UPF & UPFIO

Unbalanced Power Flow. UPFIO is the interface object to access the UPF engine.

UPS

Unbalanced Power System. The engine for the unbalanced power flow analysis.

1.20. V

Viewer

See [Browser Viewer](#).

Viewport

A window in which ADMS displays are viewed and through which the user can interact with a display.

Specifically, a specialized form used by the Browser viewer to call up displays. Each viewport contains one or more views, each of which can show a different display or different positions within a single display. A viewport can also contain other kinds of objects besides displays, such as an instance of Internet Explorer.

Voltage Reduction

A means of achieving load reduction by reducing customer supply voltage, usually by three or five percent.

1.21. Z

Zoom

The feature of simple displays that allows the display to be magnified (enlarged) or de-magnified (shrunk), and that removes objects from (declutter) or adds objects to the display as the zoom factor is changed.

Zoom Percentage

The zoom factor expressed as a percentage (10% to 1000%).

2. ADMS Acronyms

This chapter lists the acronyms used in ADMS.

Table 1. Acronyms

Acronym	Meaning
AFR	Automatic Feeder Reconfiguration
AMI	Automated Metering Infrastructure
AMS	Asset Management System
AOR	Area of Responsibility
BLA	Bus Load Allocation A Distribution Network Analysis Function
CAIDI	Customer Average Interruption Duration Index
CAIFI	Customer Average Interruption Frequency Index
CIM	Common Information Model
CIS	Customer Information System
CMI	Customer Minutes Interrupted
CMS	Customer Management System
CSS	Customer Support System
DMS	Distribution Management System
DNAF	Distribution Network Analysis Function
DNOM	Distribution Network Object Model
DOAN	Distribution Operations ANalysis
DOTS	Distribution Operations Training Simulator
DPF	Distribution Power Flow A Distribution Network Analysis Function
EMS	Energy Management System (Transmission)
ETR	Estimated Time for Restoration
FA	Fault Analysis (Short Circuit Analysis)
FISR	Fault Isolation and Service Restoration A Distribution Network Analysis Function
FISRCA	Fault Isolation and Service Restoration Contingency Analysis A Distribution Network Analysis Function
ADMS	Advanced Distribution Management System
FL	Fault Location A Distribution Network Analysis Function
GIS	Geographic Information System The GIS provides map data for the ADMS.
GUI	Graphical User Interface
GUID	Global Unique IDentifier

Acronym	Meaning
HDR	Historical Data Recording (SCADA Function)
ICCP	Intercontrol Center Communications Protocol
IEC	International Electrotechnical Commission
IVR	Interactive Voice Response System
LA	Loss Analysis A Distribution Network Analysis Function
LIM	LIMit Monitor A Distribution Network Analysis Function
LVM	Load and Volt/Var Management
MAIFI	Momentary Average Interruption Frequency Index Total number of momentary interruptions in a year/total number of consumers
NMS	Network Management System (Distribution)
NTP	Network Topology Processor
OMS	Outage Management System
PLOS	Planned Outage Study Formerly Planned Outage Request Study/Switch Orders Creation (PLORS/SOC)
PLORS	PLanned Outage Request Study
PLORS/SOC	PLanned Outage Request Study and Switch Order Creation
PQA	Power Quality Analysis
PQI	Power Quality Index Analysis
PRV	PRotection Validation A Distribution Network Analysis Function Formerly Short Circuit (SC) and Fault Analysis (FA)
RSO	Restoration Switching Order
SAIDI	System Average Interruption Duration Index Sum of outage CMIs/total number of customers served
SAIFI	System Average Interruption Frequency Index Sum of outage CIs/total number of customers served
SCADA	Supervisory Control and Data Acquisition
SOA	Service-Oriented Architecture
TC	Transfer Capacity
TCUL	Tap Changing Under Load
UI	User Interface
URD	Underground Residential Distribution
VRU	Voice Response Unit
VVC	Voltage/VAR Control (now called Load and Volt/Var Management)
WMS	Work Management System